

TAKlite User Guide

Lightweight WireGuard-hosted TAK relay, TLS CoT, datapackages, and connection user portal.

Purpose

TAKlite is a lightweight TAK relay and datapackage service for small ATAK/WinTAK teams over WireGuard VPN.

It provides:

- WireGuard VPN access for admins and TAK clients
- WGDashboard for VPN peer management
- TAKlite admin dashboard for TAK users, clients, and datapackages
- TLS CoT TCP for ATAK/WinTAK on `10.66.66.1:8089`
- Plain CoT TCP test port on `10.66.66.1:58087`
- Marti-style datapackage upload, search, download, and delete
- Connection User portal for downloading ATAK/WinTAK `.dp.zip` certificate bundles

Network Model

TAKlite is designed to be VPN-first.

Public internet:

- `51820/udp` WireGuard only
- Temporary `22/tcp` SSH only during setup, then close or restrict it

VPN-only:

- `22/tcp` SSH
- `10086/tcp` WGDashboard
- `8080/tcp` TAKlite admin UI, client portal, and HTTP datapackage API
- `8443/tcp` TAKlite HTTPS datapackage API
- `58087/tcp` plain CoT TCP
- `8089/tcp` TLS CoT TCP

Per-Install Security

Every fresh VPS installation generates new security material.

Do not copy `/root/taklite-admin`, `/etc/wireguard`, `taklite/certs`, `taklite/data`, or an old `.env` from one VPS to another.

Generated per install:

- WireGuard server private key
- Initial admin WireGuard peer key and preshared key
- WGDashboard password
- TAKlite bootstrap token
- TAKlite certificate password
- TAKlite local certificate authority
- TAKlite server certificate
- Generated ATA/WinTAK user certificates

The installer saves root-only recovery notes here:

```
/root/taklite-admin/README.txt
```

Fresh VPS Install

Create a fresh Ubuntu 26.04 LTS x64 VPS.

Option A, deploy from a GitHub clone:

```
git clone https://github.com/YOUR_GITHUB_USERNAME/taklite.git
cd taklite
scp -r . root@YOUR_VPS_PUBLIC_IP:/root/taklite
```

On the VPS:

```
cd /root/taklite
chmod +x install.sh smoke-test.sh
./install.sh
```

Option B, deploy from a release zip:

```
scp taklite-vps-bundle.zip root@YOUR_VPS_PUBLIC_IP:/root/
```

On the VPS:

```
cd /root
unzip taklite-vps-bundle.zip
cd /root/taklite-vps-bundle
chmod +x install.sh smoke-test.sh
./install.sh
```

Use the defaults unless the VPS environment requires different values.

Important defaults:

```
WireGuard server: 10.66.66.1
```

```
Admin VPN peer: 10.66.66.2
WGDashboard: http://10.66.66.1:10086
TAKlite admin: http://10.66.66.1:8080/
Client portal: http://10.66.66.1:8080/connect/
TLS CoT: 10.66.66.1:8089
Plain CoT: 10.66.66.1:58087
Certificate password: generated per install and saved in `/root/taklite-admin/README.txt`
```

Save Install Output

At the end of installation, save:

- Admin WireGuard `.conf`
- Admin WireGuard QR path
- WGDashboard username/password
- TAKlite bootstrap token
- TAKlite admin URL

Recovery files are saved on the VPS:

```
/root/taklite-admin/README.txt
/root/taklite-admin/bootstrap-token.txt
/root/taklite-admin/admin-wg0.conf
/root/taklite-admin/admin-wg0.png
```

If needed:

```
cat /root/taklite-admin/README.txt
```

Admin VPN Login

Pull the admin WireGuard config:

```
scp root@YOUR_VPS_PUBLIC_IP:/root/taklite-admin/admin-wg0.conf .
```

Import `admin-wg0.conf` into WireGuard and connect.

Open:

```
WGDashboard: http://10.66.66.1:10086
TAKlite:      http://10.66.66.1:8080/
```

First TAKlite Admin Account

Open:

```
http://10.66.66.1:8080/
```

Use the bootstrap token from install output or:

```
cat /root/taklite-admin/bootstrap-token.txt
```

Create the first TAKlite admin username/password.

Create A TAK Device User

Use this workflow for each ATAK/WinTAK device.

1. In WGDashboard, create a WireGuard peer for the user device.
2. Give the user the WireGuard QR or `.conf`.
3. In TAKlite, open Connection Users.
4. Enter a username, temporary password, and optional note.
5. Click Create User.
6. The user appears in the Connection Users table.
7. Use Download DP.zip, QR, or Copy URL.

Connection User actions:

- Download DP.zip: admin downloads the user's ATAK/WinTAK cert bundle
- Copy URL: copies the VPN-only portal URL
- QR: shows a QR for the user's portal page
- Edit: updates display name and note
- Reset Password: changes the user's portal password
- Allow/Disable Re-download: controls repeated downloads
- Reissue: revokes the old cert bundle and creates a fresh one
- Revoke: disables the user and cert package
- Delete: removes the user record, with optional generated file deletion

User Device Flow

For the end user:

1. Scan/import the WireGuard config.
2. Connect WireGuard.
3. Open the TAKlite portal QR or URL:

```
http://10.66.66.1:8080/connect/
```

4. Log in with the username/password from the admin.
5. Download the `.dp.zip`.
6. Import the `.dp.zip` into ATAK or WinTAK.
7. If prompted, use the certificate password from:

/root/taklite-admin/README.txt

The imported package configures:

Host: 10.66.66.1
Port: 8089
Protocol: SSL/TLS

ATAK Test

On Android:

1. Confirm WireGuard handshake is active.
2. Import the `.dp.zip` in ATAK.
3. Confirm the server connection turns green.
4. Drop a point.
5. Send chat.
6. Send a datapackage.
7. Confirm other clients receive the point/chat/datapackage.

WinTAK Test

On Windows:

1. Confirm WireGuard is connected.
2. Import the `.dp.zip` in WinTAK.
3. Confirm the server connection turns green.
4. Drop a point.
5. Send chat.
6. Send a datapackage.
7. Confirm ATAK receives it.

Datapackage Admin

The TAKlite dashboard shows uploaded datapackages.

Admin actions:

- Download datapackage to local machine
- Delete datapackage record and stored file

Expected API activity in logs:

```
POST /Marti/sync/missionupload
GET /Marti/sync/search
GET /Marti/sync/content?hash=...
```

Connected Clients

The dashboard shows:

- Callsign/name
- UID
- IP address
- Transport mode
- Client certificate common name
- Connection uptime
- Connected time
- Last seen time

TLS clients should show a certificate common name matching the created Connection User or package name.

Troubleshooting

Watch logs:

```
cd /root/taklite-vps-bundle
docker logs -f taklite
```

Check services:

```
docker compose ps
curl -fsS http://10.66.66.1:8080/api/health
wg show
```

Check ports:

```
ss -tulpn | grep -E '(:8080|:8443|:58087|:8089|:10086|:51820)'
```

Common issues:

- Portal login invalid: confirm the user was created under Connection Users, not Connection Packages.
- Invalid truststore: create a fresh Connection User after deploy and import the newest `.dp.zip`.
- Red TAK connection: confirm VPN is connected, use `10.66.66.1:8089:ssl`, and watch logs for `cert_cn=...`.
- Datapackage send fails: confirm the TAK client has the server connection from the `.dp.zip` and can reach `10.66.66.1:8080` or `10.66.66.1:8443` over VPN.

Smoke Test

From the VPS:

```
cd /root/taklite-vps-bundle  
./smoke-test.sh 10.66.66.1 TAKLITE_BOOTSTRAP_TOKEN_OR_ADMIN_TOKEN
```

Routine Admin Checklist

1. Connect admin WireGuard.
2. Open WGDashboard.
3. Open TAKlite admin dashboard.
4. Create WireGuard peer for user.
5. Create TAKlite Connection User.
6. Give user WireGuard QR.
7. Give user portal QR/link and password.
8. User downloads/imports `.dp.zip`.
9. Confirm client appears in Connected Clients.
10. Test PLI, chat, marker drops, and datapackages.